SUBMISSION REVIEW / 9 NOVEMBER 2026

Submission review and experimental roadmap

Measuring Advice-Style Language in OLMo-2

A promising measurement case study, but not yet a convincing top-tier empirical submission. The writing can be improved now; independent human validation and a fresh, adequately budgeted checkpoint evaluation remain substantive evidence gaps.

The contribution worth building around

The strongest question is: when we count advice-style language, how much does our conclusion depend on the detector and on how much of the answer we inspect? This is more defensible than claiming to measure empathy, identify the training source of a behavior, or explain the causal effect of DPO.

The current results offer a concrete warning. In recovered final-7B outputs, the advice-minus-emotional-control empathy-marker gap is -14.7 percentage points over the complete cached text but +25.3 points over its first 64 tokens. For structuring, the advice-minus-neutral-how-to gap is +39.0 points with exact matching and -3.0 points with the exact-or-embedding rule. These are sizeable changes in a measurement conclusion. The full-window empathy comparison is uncertain; a change in point-estimate sign is not proof of two opposite population effects.

The general facts that length matters and automatic metrics need validation are already established. The distinctive contribution should therefore be the controlled, reproducible demonstration of how detector choice and observed text interact with condition and checkpoint comparisons, together with a practical validation procedure. The present one-checkpoint recovery is useful evidence, but broad generalization needs the prospective study.

A successful paper need not obtain uniformly positive or statistically significant results. A well-estimated failure of a detector, a robust measurement interaction, or a carefully bounded null can be the main result. Select analyses for the question they answer, not for whether they improve the apparent result.

01 / EVIDENCE AUDIT

What the existing results establish

Detector diagnostic

120 AI-labeled sentences from 97 responses. Union precision is about .22, .33, .06, and .49 for the four categories; corresponding true-positive counts are only 2, 1, 1, and 26. Referral annotation is narrower than the broad referral/disclaimer rule. This is evidence about agreement with these AI labels, not human construct validity. Exact matching has no observed false positives here, but the sample is too small to establish generally high precision.

Recovered output sensitivity

700 final-7B responses support comparisons of detector components and 64/128-token windows. They do not supply a fresh test set or recover the missing raw outputs for other checkpoints. The original 256-token cap matters: the reported cap proxy is .880 on advice, .573 on factual, and .960-.980 on targeted controls. Complete cached text is often an incomplete answer.

Historical checkpoint comparisons

Under the historical rule, SFT-to-DPO validation gaps increase by .213 (1B) and .147 (7B), while structuring gaps decrease by .367 and .277. These same-interface comparisons describe detector outputs. Base-to-SFT additionally changes input formatting. Missing stagewise raw outputs prevent retrospective validation of the trajectory with the revised checks.

DPO mixture associations

The paired analysis covers 378,301 1B and 378,341 7B pairs. Length-conditioned broad-referral odds ratios are 1.088 and 1.134; structuring odds ratios are 1.128 and 1.190. These are associations in preference data, not changes in model behavior or choice probability. Narrower-referral intervals include one, which does not establish equivalence or absence of an effect.

Dependence and replication

The two DPO mixtures share 21,508 content-identical pairs. Treat the two sizes as related analyses, not fully independent replications. Prompt-cluster uncertainty and one-pair-per-prompt checks are appropriate strengths. Corpus string occurrence is supporting context, not training-data attribution.

Prospective evidence

The repository contains substantial preparation: frozen model identities, matched scenario families, a calibration design, blinded annotation tools, and generation/analysis code. Preparation and passing software checks are not empirical completion. Genuine human labels, calibration, and the full six-checkpoint evaluation are still pending in the inspected evidence.

02 / EXPERIMENTAL STRUCTURE

Three questions, three experiments

Use three research questions and give each experiment an explicit target quantity, analysis unit, comparison, and interpretation. Keep historical diagnostics separate from prospective results. Reuse the existing protocol and implementation; do not restart the project or silently modify frozen designs.

Completed during this review: direct detector comparison

A new retrospective analysis on 9 November directly estimates the exact-minus-union change in the advice-minus-targeted-control gap. Approximate 95% simultaneous intervals cover all four contrasts; values below are percentage points.

Category / targeted control	Change	95% interval
Empathy / emotional control	+47.0	[+28.1, +65.9]
Validation / emotional control	-9.7	[-22.7, +3.3]
Referral/disclaimer / health facts	+6.7	[-14.7, +28.0]
Structure / neutral how-to	+42.0	[+21.9, +62.1]

20,000 shared bootstrap draws resample prompts within condition, retaining all five seeds and both detector components. All 40 original rate cells match. This strengthens retrospective detector-choice sensitivity for empathy and structure; it is neither human validation nor fresh replication. Direct inference on prefix-window changes remains pending.

RQ1: How sensitive is the measurement?

Extend the completed exact-versus-union comparison to embedding-only rules and fixed 64/128-token prefixes, then repeat it prospectively. For each category, estimate the change in the advice-minus-matched-control gap across rules/windows directly. Preserve pairing across rules, windows, seeds, and checkpoints. Report percentage-point effects with cluster intervals; significance in one window but not another is not itself a test of change.

RQ2: Which condition gaps change across checkpoints?

Use the prepared 40 held-out scenario families with four variants each, six SFT/DPO/Instruct checkpoints across 1B/7B, and three seeds: 2,880 responses. The emotional-framing and request manipulations occur within a family, with domains balanced. Analyze family-level contrasts and changes in those contrasts across stages. Resample whole families with every dependent variant, seed, and checkpoint retained together. The independent scenario count is 40, not 2,880.

RQ3: How are marker matches associated with DPO preferences?

Retain the completed paired conditional analysis as a secondary explanation of phrase occurrence in the training mixtures. Show absolute paired differences alongside odds ratios and length distributions. Keep the spline model, source heterogeneity, exact-prompt dependence, and cross-mixture overlap visible. No additional retraining is needed for this observational question; retraining would be needed for a causal attribution question.

First: validate and generate

Priority 1 - Align and validate the labels

Separate an empathy expression from a phrase resembling one; separate emotional validation, an actual recommendation to contact a professional, a disclaimer, and organizational language. The existing broad detector can remain as a historical baseline. Use two distinct blinded human annotators and adjudication, with the prompt available as context. Report category-specific agreement, disagreements, precision/recall, support counts, uncertainty, and performance by checkpoint/condition/length where estimable. Do not replace human labels with additional AI agreement.

Priority 2 - Complete development calibration before evaluation

The prepared design includes 96 responses spanning all six checkpoints, with 1,024 and 2,048 tokens as provisional primary/sensitivity caps. The earlier eight-response 1B probe does not demonstrate adequate budgets for the other checkpoints. Verify actual stop reasons, inspect semantic completeness, time annotation and generation, and preserve any revised design as a new version. Keep all evaluation outputs unseen while choosing rules, thresholds, and sampling decisions.

Priority 3 - Run the fresh matched study

Complete human review of the drafted scenario manipulations, then generate the 2,880 primary responses and the pre-specified 192-response longer-budget subset. Retain raw text, seeds, revisions, exact serialized inputs, output-token counts, and stop reasons. A prefix comparison changes the observation window; a matched longer-budget run checks whether generation truncation itself changes the conclusion. These are different diagnostics.

Extensions if the claim requires them

Base-to-SFT training effect

If retaining a claim about SFT training itself, add a checkpoint-by-input-format crossover with the same prompts and scoring. Otherwise keep Base-to-SFT as a stage-plus-interface descriptive comparison or move it to the appendix.

General behavior across models

A second independently trained model family would strengthen external validity. It is valuable for a broad claim about LLMs; it is not a substitute for validating the OLMo-2 measurement first. Keep an OLMo-2 case-study title if the evaluation remains in one family.

Then: estimate and report

Priority 4 - Use a defensible validation sample

The prepared probability sample has 624 responses across 48 cells, with up to 120 additional detector-positive items for diagnosis; the combined workload is 624-744 unique responses, each independently labeled twice. Estimate prevalence and recall from the probability sample using its inclusion probabilities. Keep positive-enriched examples separate from population estimates. Full-response labels do not automatically validate clipped prefixes; label prefixes separately if making semantic claims about them.

Priority 5 - Plan precision, not just a response total

The current counts are planning quantities, not a demonstrated power calculation. Before seeing evaluation results, define a scientifically meaningful change and use development data or a sensitivity simulation over plausible family correlations and category rates to assess interval widths and detection probability. Prioritize adequate numbers of independent families and positive labels. If rare categories remain imprecise, report that limitation rather than changing endpoints after inspecting the result.

Priority 6 - Make the main result visual and auditable

Show detector/window gaps with uncertainty in one main figure or compact table, then show family-level checkpoint interactions. Put giant condition-rate tables, every source split, corpus counts, and redundant robustness details in the appendix. Release the exact inputs and analysis that produced every main result. Use one result manifest to keep tables and prose synchronized.

Extensions if the claim requires them

Training examples cause the behavior

This requires interventions such as matched data removal or label changes followed by retraining, with random/source/length-matched controls and general-capability checks. Existing corpus counts and conditional DPO models cannot establish this. Treat this as a separate, resource-heavy project unless causal attribution is central.

A better detector

If introducing a calibrated detector, train or tune it on development families, lock it before evaluation, and compare against the historical exact and union baselines using independent human test labels. Improving the detector on the existing audit alone would produce an in-sample correction, not evidence of generalization.

05 / READABILITY

Make the paper easy to follow

Open with the question and a short example, then state what was measured. Define a marker as a textual feature and a detector as a matching rule once. Use category names consistently. The historical name empathy-opener must not imply the algorithm searches only the beginning; it searches all eligible sentences.

Write each results paragraph in the order question, estimate, uncertainty, interpretation. Prefer 'the gap increased by 21.3 percentage points' to a chain of unexplained decimals. Define the comparison direction and control in every table caption. Explain that OR=1 is the no-association reference, and an OR of 1.13 is not a 13-percentage-point preference increase.

Keep one local caveat where it changes interpretation, then collect broader limits in Limitations. Repeating every caveat in the abstract, introduction, every result, discussion, and conclusion obscures the contribution. Avoid treating non-significance as equivalence, three detector thresholds as construct validation, or AI-human agreement as interchangeable with AI-AI agreement.

The manuscript contains two slightly different final-7B sources: an archived advice empathy rate of .617 and a recovered-cache rate of .6133. This is a provenance discrepancy, not rounding. Preserve and identify the source of each result; do not silently harmonize values. Also distinguish original analyses, later exploratory fits, and new retrospective sensitivity analyses by their actual chronology.

For an eventual submission, use a compact evidence map, one main detector/window result, one checkpoint-interaction result, and one secondary DPO result. Check venue-specific length, anonymity, ethics, and artifact requirements after the venue is selected. No target venue or deadline was supplied for this review.

Use the same sequence for each result: question, estimate, uncertainty, interpretation. Report effect sizes in percentage points and define the comparison direction.

What ready for submission means

Call the scientific revision ready for a submission decision when the human targets are aligned and independently labeled, the main held-out comparison has complete raw outputs, the key detector/window and checkpoint effects have uncertainty based on their independent units, and the conclusion survives the pre-specified robustness checks or is narrowed to reflect their failure. A null result can meet this criterion; a favorable p-value alone cannot.

Have two readers unfamiliar with the project independently answer: What is the research question? What is the main finding? What can this study not establish? Use their answers to identify remaining jargon and structural problems. Their participation has not been completed by an AI review.

The current manuscript can be circulated as a clearer revision draft. It should not be described as validated, causally explanatory, or guaranteed competitive at a top venue. The most useful next investment is the prepared human-validation and matched-checkpoint study, not another round of cosmetic edits or a larger unvalidated prompt collection.

Local evidence inspected

Current manuscript - acl_latex.tex

Recovered-output diagnostics - raw_output_robustness.json

AI-label diagnostic - ai_consensus_validation_v3.json

Checkpoint contrasts - interaction_tests_v3.json

Paired DPO analysis - conditional_results.json

Prospective validation protocol - 15 September 2026

Extended study protocol - 15 September 2026

Frozen extended design - full_study/protocol_v1.json

Completion audit - 15 September 2026

New detector comparison - paired_measurement_contrasts.json (9 November 2026)

How to read the literature comparison

The following pages summarize a targeted review of ten relevant works, not a systematic literature review. Verified methods and proposed adaptations are separated. Lee et al. was verified from its repository abstract only; detailed implementation claims are withheld. Ibrahim et al. is cited as the 2026 journal paper, with detailed methods checked in the earlier preprint.

Metric gains and human judgments

1. Lee, Yoon Kyung; Lee, Jaehyeong; An, Seonu; Hahn, Sowon (2026).

What Makes an LLM Response Read as Empathic? A Multi-Dimensional Study of Empathic Alignment.

Source: Primary source

Verification: repository abstract/metadata only; full PDF inaccessible in this audit. Do not infer sample sizes, exact model IDs, training settings, statistical tests, or code availability.

Verified method: five instruction-tuned LLMs assessed on specificity, emotional reflection, affective word choice, and diversity. Reflection-targeted preference learning improved the four-dimensional metric composite, but human raters showed no reliable preference over baseline responses.

Overlap/difference: closest missing warning that optimizing empathy-related measurements need not improve perceived response quality. Your study examines measurement instability across checkpoints and text windows rather than optimizing a model.

Proposed adaptation: compare marker changes with separately collected contextual human judgments, reporting disagreement rather than assuming convergence.

2. Kumar, Aakriti; Poungpeth, Nalin; Yang, Diyi; Farrell, Erina; Lambert, Bruce L.; Groh, Matthew (2026).

When large language models are reliable for judging empathic communication. Nature Machine Intelligence 8, 173-185.

Published source: Primary source

Full-text methods checked: Primary source

Verified method: 200 conversations, 50 from each of four datasets/frameworks, with 3,150 expert, 2,844 crowd, and 3,150 LLM annotations. Models include GPT-4o, Gemini 2.5 Pro and Claude 3.7 Sonnet. Reliability is assessed against expert agreement across framework subcomponents, rather than assuming a universal empathy label. Models can approach expert reliability in these validated settings.

Code: Code repository (opened; contains annotations and analysis.ipynb).

Overlap/difference: important counterbalance to a categorical claim that LLM empathy judges are unreliable; your narrow AI-only audit does not establish validity or invalidity for all such judges.

Proposed adaptation: report human-human and detector-human agreement separately, with a declared response/context unit and per-category errors.

Style and construct validity

3. Ibrahim, Lujain; Hafner, Franziska Sofia; Rocher, Luc (2026).

Training language models to be warm can reduce accuracy and increase sycophancy. Nature 652, 1159-1165.

Published record: Primary source

Author source: Primary source

Full-text methods checked in earlier preprint: Primary source (title differs; do not assume every detail is unchanged in final journal version).

Verified preprint method: five models, warmth-oriented SFT on transformed conversational responses; LoRA for open models. Evaluations pair original and warm models on factual/reliability tasks and vary emotional context and incorrect user beliefs. Controls include cold-style fine-tuning, system prompting, response length, and general capabilities. Analyses include McNemar tests, FDR correction, and logistic regression.

Overlap/difference: establishes why style measurement matters, but studies downstream correctness and sycophancy under interventions; your study does not estimate those outcomes.

Proposed adaptation: use matched scenario variants and distinguish acknowledgment of emotion from agreement with an incorrect proposition. Do not add a large safety claim without measuring safety.

4. Bean, Andrew M., et al. (2025).

Measuring what Matters: Construct Validity in Large Language Model Benchmarks. NeurIPS 2025 Datasets and Benchmarks track.

Sources: Primary source ; Full text

Verified method: systematic review of 445 benchmarks by 29 expert reviewers, selected from proceedings using inclusion/exclusion criteria and a structured codebook. Separates phenomenon, task, metric, and claim; provides eight recommendations and a construct-validity checklist.

Code: Code repository (opened).

Overlap/difference: establishes the broader methodological problem. Your contribution would be empirical quantification of one concrete failure mode, not a new general account of construct validity.

Proposed adaptation: make a small construct-to-operation table specifying intended phenomenon, observed marker, unit, likely confounds, validation target, and permitted interpretation. Use a frozen held-out validation split and an explicit error analysis.

Training stages and attribution

5. Rivière, Pamela D.; Jones, Cameron; Trott, Sean (2026).

Developmental Trajectories of Situation Modeling and Mentalizing in Transformer Language Models.

Source: Primary source

Verified method: OLMo-2 and Pythia checkpoint trajectories; 192 passages from 12 false-belief templates varying knowledge state, cue, and mention order. Adds 192 antagonist variants and 768 situation-modeling queries. Uses correct/distractor completion log-odds, Hugging Face Transformers, log-spaced checkpoints and targeted stress tests.

Paper-linked resources: Code repository ; OSF materials (paper links verified; repository content was not accessible).

Overlap/difference: close model-family and stage framing, but structured reasoning choices rather than free-form advice markers.

Proposed adaptation: make variants within each scenario family, preserve family IDs across stages, and test the relevant stage-by-condition contrast. Their token-probability endpoint avoids free-generation length confounding; it is not interchangeable with your occurrence endpoint.

Bibliographic status: existing citation is real and accurate; arXiv describes a non-archival workshop submission.

6. Xiao, Frank; Aranguri, Santiago (2026).

Probe-Based Data Attribution: Discovering and Mitigating Undesirable Behaviors in LLM Post-Training (explicitly v3).

Source: Primary source

Verified method: activation-difference vectors for behavior changes and preference pairs, ranked by cosine similarity. OLMo-2 7B DPO interventions filter or relabel ranked data and retrain from SFT; random, gradient and LLM-based rankings serve as comparisons. Behavioral evaluation includes 120 prompts with 100 response samples each and prompt-bootstrap intervals; capability checks assess trade-offs.

Overlap/difference: direct precedent on the same post-training chain. It validates data attribution through retraining; corpus phrase co-occurrence and conditional DPO association cannot supply that evidence.

Proposed adaptation: retain observational wording. If attribution becomes the central claim later, add targeted-data removal/relabeling versus size-matched random controls, held-out prompts and capability checks; this is a separate substantial experiment.

Bibliographic status: current v3 title in the manuscript is correct and is also displayed by the unversioned arXiv page. Earlier versions used an In-the-Wild Model Organisms title. A versioned URL makes the cited version explicit.

Response length and evaluation

7. Dubois, Yann; Galambosi, Balázs; Liang, Percy; Hashimoto, Tatsunori B. (2024; revised 2025).

Length-Controlled AlpacaEval: A Simple Way to Debias Automatic Evaluators.

Source: Primary source

Verified method: on 805 instructions, fit a logistic preference model with model identity, transformed length difference and instruction difficulty, then predict with zero length difference. Evaluates resistance to verbosity manipulations and correlation with external human rankings.

Code: Code repository (opened).

Overlap/difference: establishes length-sensitive preference evaluation. Your endpoint is detector-defined marker presence, and prefix restriction changes how much text is observed rather than estimating the same quantity as a preference-regression adjustment.

Proposed adaptation: label full-response, fixed-prefix and conditional-model endpoints separately; publish all of them. Do not describe prefix truncation as a causal removal of length bias or equate conditional association with a training effect.

8. Singhal, Prasann; Goyal, Tanya; Xu, Jiacheng; Durrett, Greg (2023; revised 2024).

A Long Way to Go: Investigating Length Correlations in RLHF.

Source: Primary source

Verified method: LLaMA-7B experiments across WebGPT, Stack Exchange and RLCD settings, using Hugging Face TRL and rank-16 LoRA. Examines length-stratified reward changes, length-only reward, PPO/reward-model interventions, and supplementary DPO experiments. Reports downstream simulated-preference evaluations alongside reward and length.

Overlap/difference: length correlations in post-training are established. The paper intervenes on training; your marker/window analysis asks a narrower measurement question.

Proposed adaptation: report response-length distributions and truncation rates alongside marker rates; treat length-only or sentence-count baselines as diagnostic controls. More tokens or more marker hits do not by themselves establish better advice.

Human evaluation and judge reliability

9. Welivita, Anuradha; Zeitoun, Fawzia; Pu, Pearl (2026).

Who Generates More Empathetic Responses - Humans or LLMs? A Comparative Evaluation with Human and LLM Judges. CoNLL, 358-381.

Sources: Primary source ; Full text

Verified method: 2,000 dialogue prompts across 32 emotions; four LLMs and a human baseline; 1,000 participants in a between-subjects study with source-blinded ratings. GPT-4o-mini supplies an additional evaluation. Human groups receive the same empathy explanation as generation prompts; ratings are Bad/Okay/Good and analysis uses chi-square comparisons. The limitations explicitly state response length was reported but not controlled or analyzed.

Overlap/difference: human-perceived response quality rather than lexical-marker incidence; the length limitation motivates a targeted extension but does not prove its findings are artifacts.

Proposed adaptation: collect contextual human labels on a source-blinded sample, include output length in the analysis plan, and use cluster-aware comparisons for the repeated scenario structure of this study.

10. Badawi, Abeer; Rahimi, Elahe; Laskar, Md Tahmid Rahman; Grach, Sheri; Bertrand, Lindsay; Danok, Lames; Dhanesh, Prathiba; Huang, Jimmy Xiangji; Rudzicz, Frank; Dolatabadi, Elham (2026).

When Can We Trust LLMs in Mental Health? Large-Scale Benchmarks for Reliable LLM Evaluation. EACL, 3873-3896.

Sources: Primary source ; Full text

Verified method: MentalBench combines 10,000 conversations with human and nine model responses. MentalAlign samples 1,000 conversations; four LLM judges and three human experts assess seven attributes. Uses ICC for consistency/absolute agreement, bias estimates, and bootstrap intervals. Its ICC analysis aggregates over conversations to model-level scores; it should not be treated as response-level classification accuracy.

Code/data: Code repository (opened).

Overlap/difference: task-specific evidence that agreement varies by attribute; does not justify a universal claim that automated judges fail.

Proposed adaptation: validate each marker category separately and report false-positive/false-negative patterns; reserve semantic empathy claims for an appropriately validated contextual label.

Where this paper can contribute

The established literature already addresses style changes, length-sensitive evaluation and the need to validate metrics. The distinctive contribution here should be a reproducible, human-checked demonstration of how detector components and observed response windows alter the same checkpoint comparison. Keep the title and conclusions at the scope supported by that evidence.